Dual-Mode Automated Window Blinds Control System

Mechatronics project integrating autonomous and manual control of window blinds.

Hunter Fuller

huntermfuller@email.com hunter-m-fuller.github.io
linkedin.com/in/hunterfuller

OBJECTIVES

- Window blinds require constant manual adjustment to balance light and privacy throughout the day.
- Design a system that responds to ambient light automatically, with optional IR remote override for manual control.

OUTCOMES & CONTRIBUTIONS

- Built a fully functional wall-mounted enclosure housing all electronics, 3D modeled and printed in-house.
- Integrated dual-input control — photoresistors for autonomous mode and IR remote for manual — driving a DC stepper motor to position the blinds.
- Delivered clear system feedback through an RGB LED, passive buzzer, and LCD screen.

TECHNICAL DETAILS & SKILLS

- Arduino programming for state-based motor control, photoresistor comparison logic, and IR signal parsing.
- Circuitry design incorporating a stepper motor, photoresistors, IR receiver, passive buzzer, RGB LED, and LCD screen.

Arduino **SolidWorks** **Circuitry** **Embedded Systems**

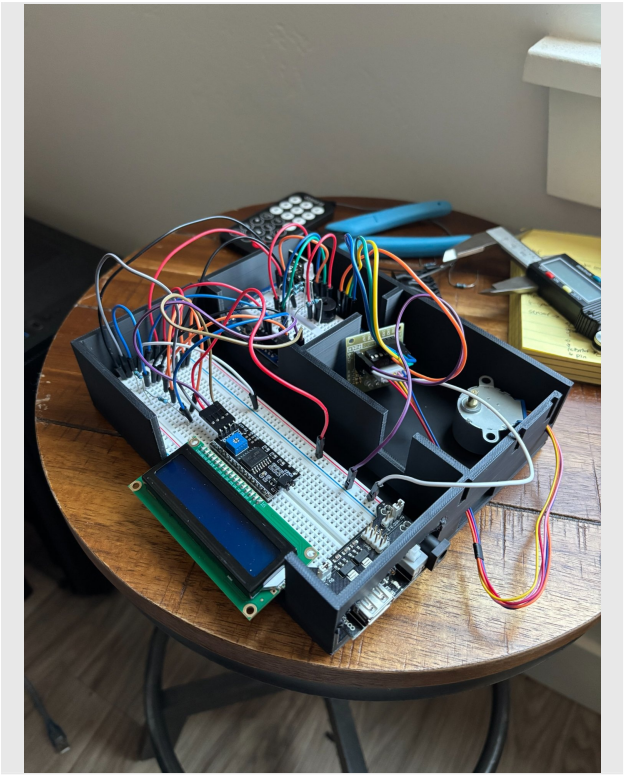

Fig. 1 — Electronics integrated into the 3D-printed enclosure during assembly, showing the LCD screen, passive buzzer, RGB LED, IR sensor, photoresistor (top left), and DC stepper motor (top right).

Fig. 2 — Completed enclosure powered on, LCD displaying “MANUAL MODE” and RGB LED indicating system status.

Fig. 3 — Final product mounted on the wall adjacent to the window blinds, stepper motor coupled to the blind rod.